Exam 1
CpE 3150-Digital System Design

Name:_____

Student #:_____

This exam is an individual activity. The exam is open textbook (Mazidi textbook and ATmega32 datasheet) and other class materials posted on Canvas (notes, videos, homework solutions, quiz solutions, old exams, Microchip Studio workspaces) for this course. Calculators are permitted (just to be clear). Other outside materials are NOT permitted. Answers given that are identified as coming from outside sources or as shared work with another classmate will treated as academic dishonesty. Any problems with answers identified as dishonest will result in an exam score of 0 points, and there may be a referral to the ECE Department Chair for academic dishonesty.

Show all work. Partial credit will be given based on work shown.

I completed this exam on my own following the guidelines above and did not obtain or provide assistance to anyone else taking this exam. This includes not seeking online assistance from others.

_____ _____

Signature Date

(70 points) Part 1 of the Exam is a quiz (Exam 1: Part 1) on Canvas. This part of the exam consists of True/False, Multiple Choice, and Matching problems. No work is required to be shown for this part of the exam. You may submit any work for this part of the exam on separate sheet(s) with the problem(s) labeled.

(**30 points**) Part 2 of the Exam is given as follows. Showing your work is necessary for partial credit. You may print out this part of the exam and write your solutions, scan them, and submit the scanned document on Canvas for Exam 1. You may also provide your solutions (with work shown) in this Word document and submit on Canvas.

Both parts of the exam are subject to the integrity statement that you are signing above. Both parts of the exam must be completed (part 1 on Canvas) and submitted on Canvas (part 2 here) by the exam deadline. Best wishes with the exam!

Exam 1: Part 2

1. For the ATmega32 program below, answer the following questions.

Number of Machine Cycles for Each Instruction

Instruction	Number of Cycles	Instruction	Number of Cycles
LDI	1	RET	4
IN	1	NOP	1
OUT	1	DEC	1
CALL	4	BRNE	2/1
COM	1	RJMP	2

```
.ORG 0
        LDI R16,HIGH(RAMEND)
        OUT SPH,R16
        LDI R16,
LOW(RAMEND)
        OUT SPL,R16


LOOP:
LDI R16, 0xFF
OUT PORTD, R16
LDI R16, 0x00
        CALL DELAY
        RJMP LOOP

.ORG 0x30
DELAY: LDI R24, 40
LOOP2: NOP
NOP
LDI R25, 50
LOOP1:  NOP
NOP
DEC R25
        BRNE LOOP1
        DEC R24
        BRNE LOOP2
        RET
```

(10 Points) Determine the **total machine cycles** AND **time delay** (in **microseconds**) for the instructions in the DELAY subroutine (given in gray). Assume that the crystal frequency is 16 MHz. Show your work for full credit.

LOOP1: [1 NOP+1NOP+1DEC+2BRNE]50-1BRNE = 249 MCs

LOOP2: [1NOP+1NOP+1LDI+LOOP1+1DEC+2BRNE]40 – 1BRNE
[6+249]40-1 = 10199 MCs

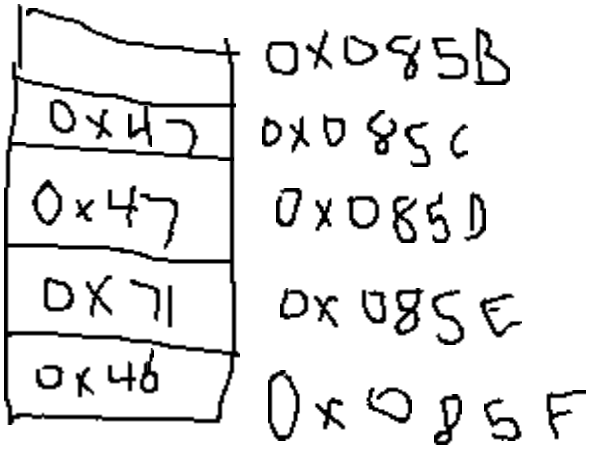
DELAY: 1LDI+LOOP2+4RET = 10204 MCs

tdelay = 10204 MCs (1cc/1MC)(1s/16 10^6 cc) = 637.75 10^-6 s = 637.75 us

2. **(10 Points)** Given the following sequence of code. Determine the values of the SP and registers in **hex** specified **after** each instruction is executed. The initial value of SP for this program sequence is SP = 0x0000 (given in the table below). Hints: 1) remember the top location of data memory for the ATmega32 and 2) the number of bits in SP and the GPRs (R20, R21, R22).

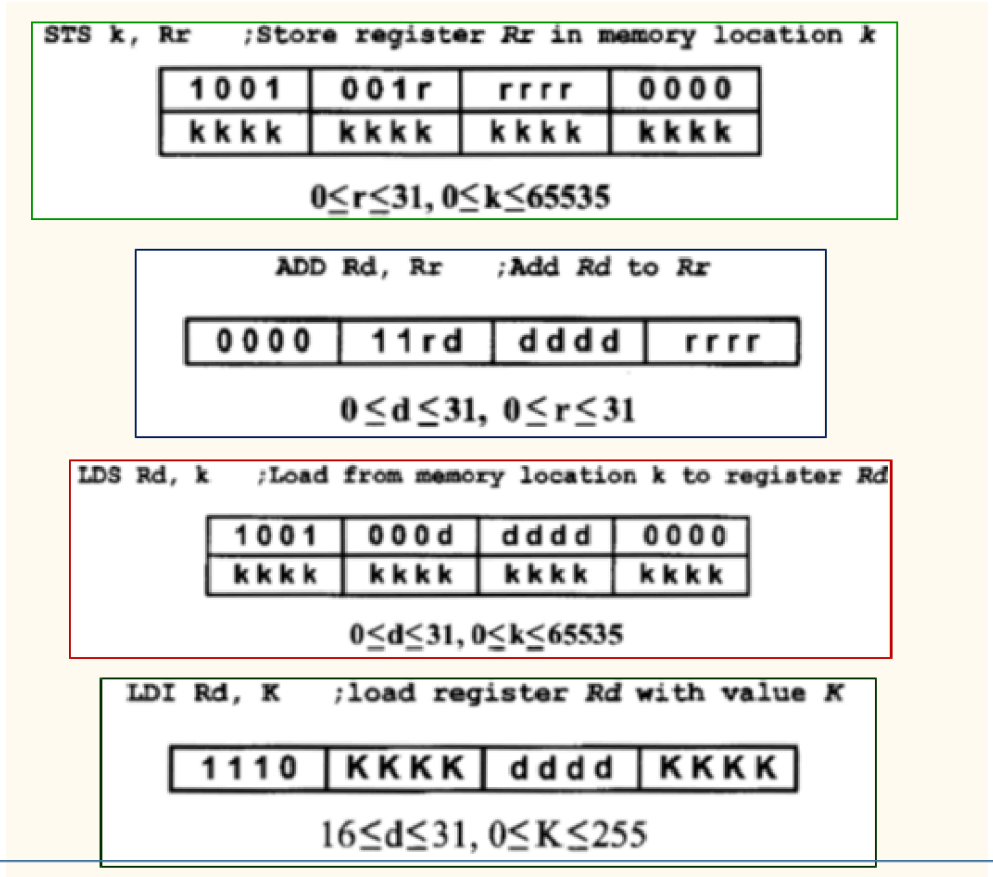
Answers in hex

	SP	R20	R21	R22
	0x0000			
LDI R16, LOW(RAMEND)				
OUT SPL, R16	0x005F			
LDI R16, HIGH(RAMEND)				
OUT SPH, R16	0X085F			
LDI R22, 71				0X47
LDI R21, $71			0X71	
LDI R20, 0b01000110		0X46		
PUSH R20	0X085E			
PUSH R21	0X085D			
PUSH R22	0X085C			
PUSH R22	0X085B			
POP R20	0X085C	0X47		
POP R21	0X085D		0X47	
POP R22	0X085E			0X71



3. (10 Points) Given the following assembly language instruction sequence and machine codes for the instructions. In the table below, fill in the starting memory location and machine code for each instruction. **Show your work for full credit**. Fill in the Memory Locations and Machine Code with **HEX** values.

Starting Code Memory Location for Instruction (**HEX**)	Machine Code (see below for different instructions) (**HEX**)	Assembly Language Program Instruction
0X00000	0X91A0 0X0910	LDS R26, 0X910
0X00002	0XE6DA	HERE: LDI R29, 0X6A
0X00003	0X0FAD	ADD R26, R29
0X00004	0XF7E9	BRNE HERE



BRNE BranchPlace; conditional branch -> branch if C = 0

1 1 1 1	0 1 K6 K5	K4 K3 K2 K1	K0 0 0 1

K6 K5 K4 K3 K2 K1 K0 = > -64 to +63 (2s complement range) offset with respect to the current PC

LDS R26, 0X910 26 => 11010

1001 0001 1010 0000
0000 1001 0001 0000 => 0X91A0 0X0910

LDI R29, 0X6A 29-16=13 => 1101
1110 0110 1101 1010 => 0XE6DA

ADD R26, R29 Rd = 11010 (R26) Rr = 11101 (R29)
0000 1111 1010 1101 => 0X0FAD

BRNE HERE
1111 0111 1110 1001 => 0XF7E9

PCNEW = PCCURRENT+k+1
0x00002=0x00004+k+1
k = -3

K6…K0
0000011 => 1111100+1 = 1111101